THE DIMENSION OF A FINITE-DIMENSIONAL SCHEME IS ATTAINED AT A CLOSED POINT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

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Lemma 0.1. *If X is a finite dimensional scheme, then there exists a closed (and hence finite type) point $x \in X$ such that $\dim_x X = \dim X$.*

Proof. Let $d = \dim X$, and choose a maximal strictly decreasing chain of irreducible closed subsets of X, say

$$Z_0 \supset Z_1 \supset \ldots \supset Z_d. \tag{0.1.1}$$

The subset Z_d is a minimal irreducible closed subset of X, and thus any point of Z_d is a generic point of Z_d. Since the underlying topological space of the scheme X is sober, we conclude that Z_d is a singleton, consisting of a single closed point $x \in X$. If U is any neighbourhood of x, then the chain

$$U \cap Z_0 \supset U \cap Z_1 \supset \ldots \supset U \cap Z_d = Z_d = \{x\}$$

is then a strictly descending chain of irreducible closed subsets of U, showing that $\dim U \geq d$. Thus we find that $\dim_x X \geq d$. The other inequality being obvious, the lemma is proved. □

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